

Advanced Statistical Methods II

Review before Midsem

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Outline of Today's Review

The whole story in one picture

Missing data review

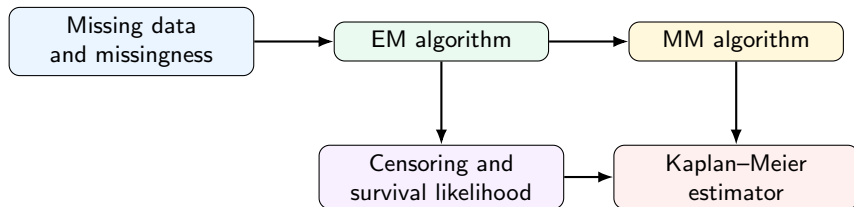
EM algorithm review

MM algorithm review

Censored likelihood review

Kaplan–Meier review

What we have studied so far



The common theme is: **observed data may not contain the whole story**. We either model what is missing, optimize through hidden structure, or estimate survival while respecting censoring.

The story so far...



Topic	Question	Main tool
Missingness	Why is a value absent?	MCAR, MAR, MNAR
Imputation	What values are plausible?	Conditional distributions
EM	How to estimate with latent data?	E-step, M-step, Q-function
MM	How to optimize a hard objective?	Surrogate functions
Censoring	What do censored observations contribute?	Likelihood using f and S
Kaplan–Meier	How to estimate $S(t)$ nonparametrically?	Product-limit estimator

Observed, missing, and complete data



We write the full data as

$$Y = (Y_{\text{obs}}, Y_{\text{mis}}),$$

where only Y_{obs} is available to us.

The **observed-data likelihood** is obtained by summing or integrating over missing values:

$$L_{\text{obs}}(\theta) = f(Y_{\text{obs}} | \theta) = \int f(Y_{\text{obs}}, Y_{\text{mis}} | \theta) dY_{\text{mis}}.$$

- ▶ If Y_{mis} is discrete, the integral becomes a sum.
- ▶ This is often the point where the observed likelihood becomes difficult.

$$R_i = \begin{cases} 1, & \text{if } Y_i \text{ is observed,} \\ 0, & \text{if } Y_i \text{ is missing.} \end{cases}$$
$$\mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi).$$

This distinction is central in missing-data analysis; see, for example, [Little and Rubin \(2019\)](#) and [van Buuren \(2018\)](#).

MCAR, MAR, MNAR: one-slide recall



Mechanism	Meaning	Typical consequence
MCAR	Missingness is unrelated to observed or unobserved data.	Complete-case analysis may be unbiased but inefficient.
MAR	Missingness may depend on observed data, but not on missing values after conditioning on observed data.	Likelihood or imputation methods can often be justified.
MNAR	Missingness still depends on missing/unobserved values.	Need explicit model for the missingness process.

EM: when and why?



EM is useful when:

- ▶ the observed likelihood is difficult;
- ▶ the complete-data likelihood is easier;
- ▶ the missing/latent part can be replaced by conditional expectations.

Examples we discussed

- ▶ missing normal observations;
- ▶ latent machine labels in a Poisson mixture;
- ▶ normal mixture models;
- ▶ incomplete multinomial or contingency tables.

The general EM algorithm was formalized by [Dempster et al. \(1977\)](#).

The Q-function



Let $X = (Y, Z)$ denote complete data, where Y is observed and Z is missing or latent.

The complete-data log-likelihood is

$$\ell_c(\theta; X) = \log f(X \mid \theta).$$

Given the current estimate $\theta^{(t)}$, define

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}} [\ell_c(\theta; X) \mid Y = y].$$

EM maximizes the expected complete-data log-likelihood, where the expectation is taken using the current parameter value.

EM algorithm: the two steps



E-step

Compute

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}} \{ \ell_c(\theta; X) \mid Y = y \}.$$

M-step

Choose

$$\theta^{(t+1)} = \arg \max_{\theta} Q(\theta \mid \theta^{(t)}).$$

Intuition: E-step fills in the missing information in expectation; M-step re-estimates parameters as if that expected complete-data information were available.

Ascent property

If the M-step satisfies

$$Q(\theta^{(t+1)} \mid \theta^{(t)}) \geq Q(\theta^{(t)} \mid \theta^{(t)}),$$

then the observed-data log-likelihood satisfies

$$\ell(\theta^{(t+1)}; y) \geq \ell(\theta^{(t)}; y).$$

EM is designed to move uphill in the observed-data likelihood, although it may move slowly and can converge to a local maximum or stationary point.

A standard mixture template



For a finite mixture,

$$f(y_i | \theta) = \sum_{k=1}^K \pi_k f_k(y_i | \theta_k), \quad \sum_{k=1}^K \pi_k = 1.$$

Introduce latent class indicators

$$Z_i \in \{1, \dots, K\}.$$

The E-step computes posterior memberships:

$$\tau_{ik}^{(t)} = \mathbb{P}_{\theta^{(t)}}(Z_i = k | y_i) = \frac{\pi_k^{(t)} f_k(y_i | \theta_k^{(t)})}{\sum_{r=1}^K \pi_r^{(t)} f_r(y_i | \theta_r^{(t)})}.$$

EM gives

$$\pi_k^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \tau_{ik}^{(t)},$$

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^n \tau_{ik}^{(t)} y_i}{\sum_{i=1}^n \tau_{ik}^{(t)}},$$

$$(\sigma_k^2)^{(t+1)} = \frac{\sum_{i=1}^n \tau_{ik}^{(t)} (y_i - \mu_k^{(t+1)})^2}{\sum_{i=1}^n \tau_{ik}^{(t)}}.$$

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Incomplete tables: the EM recipe



For multinomial or contingency-table data:

1. Write a complete table with latent cell counts.
2. Write the complete-data log-likelihood.
3. E-step: replace missing cell counts by their conditional expectations under $\theta^{(t)}$.
4. M-step: compute the usual complete-data MLEs using observed counts plus expected counts.

In many table problems, EM simply replaces missing sufficient statistics by their conditional expectations.

EM: practical warnings



- ▶ **Initialization matters:** different starting values may lead to different local maxima.
- ▶ **Slow convergence:** EM can be very slow near a maximum.

MM stands for:

- ▶ **Majorization–Minimization** for minimizing an objective;
- ▶ **Minorization–Maximization** for maximizing an objective.

For minimization of $f(\theta)$, construct a surrogate $g(\theta \mid \theta^{(t)})$ such that

$$g(\theta \mid \theta^{(t)}) \geq f(\theta), \quad g(\theta^{(t)} \mid \theta^{(t)}) = f(\theta^{(t)}).$$

Then update

$$\theta^{(t+1)} = \arg \min_{\theta} g(\theta \mid \theta^{(t)}).$$

Replace a difficult objective by an easier tangent surrogate, optimize the surrogate, and repeat.

MM descent argument



For minimization,

$$f(\theta^{(t+1)}) \leq g(\theta^{(t+1)} | \theta^{(t)}) \leq g(\theta^{(t)} | \theta^{(t)}) = f(\theta^{(t)}).$$

Step	Meaning
$f(\theta^{(t+1)}) \leq g(\theta^{(t+1)} \theta^{(t)})$	surrogate lies above objective
$g(\theta^{(t+1)} \theta^{(t)}) \leq g(\theta^{(t)} \theta^{(t)})$	surrogate minimized
$g(\theta^{(t)} \theta^{(t)}) = f(\theta^{(t)})$	tangency condition

This is the MM counterpart of the EM ascent idea.

EM and MM: same spirit, different construction



	EM	MM
Main object	Expected complete-data log-likelihood	Surrogate objective
Hidden structure	Missing or latent data	Not necessarily data-based
Guarantee	Likelihood ascent	Objective descent/ascent
Typical update	E-step, M-step	Build surrogate, optimize surrogate
Main danger	Local maxima, slow convergence	Loose surrogate, slow convergence

Think of EM as a probabilistically constructed MM-like algorithm.

MM: how to recognize a valid surrogate



For minimization, check two things:

1. **Majorization:**

$$g(\theta \mid \theta^{(t)}) \geq f(\theta) \quad \text{for all } \theta.$$

2. **Tangency:**

$$g(\theta^{(t)} \mid \theta^{(t)}) = f(\theta^{(t)}).$$

A function that is merely easier to minimize is not necessarily an MM surrogate. It must touch and dominate the target objective.

$$S(t) = \mathbb{P}(T > t) = 1 - F(t),$$

$$f(t) = F'(t),$$

$$h(t) = \lim_{\Delta \downarrow 0} \frac{\mathbb{P}(t \leq T < t + \Delta \mid T \geq t)}{\Delta} = \frac{f(t)}{S(t)}.$$

The hazard is an instantaneous event rate among those still at risk.

Exact and right-censored observations



For each subject, we observe

$$Y_i = \min(T_i, C_i), \quad \delta_i = \mathbb{I}(T_i \leq C_i),$$

where C_i is the censoring time.

Observation type	Data	Likelihood contribution
Exact event	$Y_i = t_i, \delta_i = 1$	$f(t_i)$
Right-censored	$Y_i = c_i, \delta_i = 0$	$S(c_i)$

A censored subject is not missing completely: we know the event time exceeded the censoring time.

$$L(\theta) = \prod_{i=1}^n f(y_i; \theta)^{\delta_i} S(y_i; \theta)^{1-\delta_i}.$$
$$L(\theta) = \prod_{i=1}^n h(y_i; \theta)^{\delta_i} S(y_i; \theta).$$

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Exponential example



Suppose

$$T_i \sim \text{Exponential}(\lambda), \quad f(t) = \lambda e^{-\lambda t}, \quad S(t) = e^{-\lambda t}.$$

Then

$$L(\lambda) = \prod_{i=1}^n (\lambda e^{-\lambda y_i})^{\delta_i} (e^{-\lambda y_i})^{1-\delta_i} = \lambda^{\sum_i \delta_i} e^{-\lambda \sum_i y_i}.$$

Thus

$$\ell(\lambda) = d \log \lambda - \lambda \sum_i y_i, \quad d = \sum_i \delta_i.$$

The MLE is

$$\hat{\lambda} = \frac{d}{\sum_i y_i}.$$

Let

$$t_1 < t_2 < \cdots < t_m$$

be the distinct observed event times,

At event time t_j :

- ▶ n_j : number at risk just before t_j ;
- ▶ d_j : number of events at t_j .

The risk set is

$$\mathcal{R}(t_j) = \{i : Y_i \geq t_j\}.$$

Censored observations reduce later risk sets but do not create jumps in the survival curve.

At event time t_j , estimate

$$\mathbb{P}(T > t_j \mid T \geq t_j) \approx 1 - \frac{d_j}{n_j}.$$

$$\hat{S}(t) = \prod_{t_j \leq t} \left(1 - \frac{d_j}{n_j}\right).$$

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Kaplan–Meier: how to compute by hand



Event time t_j	Risk set n_j	Events d_j	KM multiplier
t_1	n_1	d_1	$1 - d_1/n_1$
t_2	n_2	d_2	$1 - d_2/n_2$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
t_j	n_j	d_j	$1 - d_j/n_j$

Then multiply the successive factors:

$$\hat{S}(t_j) = \hat{S}(t_{j-1}) \left(1 - \frac{d_j}{n_j}\right), \quad \hat{S}(0) = 1.$$

The KM curve jumps only at event times and remains flat between event times.

The median survival time is

$$\hat{t}_{0.5} = \inf\{t : \hat{S}(t) \leq 0.5\}.$$

Interpretation:

- ▶ It is the time by which the estimated survival probability falls to 0.5.
- ▶ It is often easier to interpret than the entire survival curve, but it does not summarize everything.

A survival curve is richer than a single median; always inspect the curve and censoring pattern.

One integrated example



A medical study records time to relapse. Some patients are still relapse-free at last follow-up. In addition, a biomarker is missing for some patients, and missingness is more common among older patients.

1. The relapse times involve **right censoring**; use survival likelihood or KM methods.
2. The missing biomarker requires a **missing-data mechanism** discussion.
3. If missingness depends only on observed age and other observed variables, MAR may be plausible.
4. If a latent disease subtype is introduced, EM may be useful.

Real analyses often combine the ideas: missing covariates, latent structure, and censored outcomes can occur together.

- Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 39(1), 1–38.
- Little, R. J. A. and Rubin, D. B. (2019). *Statistical Analysis with Missing Data*. 3rd ed. Wiley.
- van Buuren, S. (2018). *Flexible Imputation of Missing Data*. 2nd ed. CRC Press.
- Kleinbaum, D. G. and Klein, M. (2012). *Survival Analysis: A Self-Learning Text*. 3rd ed. Springer.

**On 1st September (Tuesday), we
have a QUIZ.**

**On the other two lectures next week, we
shall have problem solving sessions.**